

BOTANY XI

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF BIOLOGY.

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QUESTIONS	YEARS
Chapter 3: CELL STRUCTURE AND FUNCTION	
Section B	
iv) Why chloroplast considered as an energy converting organelle? v) Write a note on Nucleus OR Mitochondria. viii) Differentiate between any One of the following: *Rough Endoplasmic Reticulum and smooth Endoplasmic Reticulum.	2025
i) Why Lysosome is called suicidal sac? ii) Why are chloroplast and mitochondria considered as semi-autonomous organelles?	2024
2. xiii) Explain the movement mechanism of Cilia and Flagella. 2. xiv) Write significance of differential staining.	2023
2. (i) Mention any two differences between prokaryotic and eukaryotic cell. 2. (xi) Name two techniques used to isolate the component of cell. 3. (i) Draw a labelled diagram of fluid mosaic model of plasma membrane. 3. (ix) Write note on any one of the following: (a) Nucleus (b) Mitochondria	2018
2. (ii) State one function each of the following: (a) Ribosome (b) Mitochondria 2. (v) Name various components of the nucleus.	2017
ii) What is the cause of lysosomal storage disease? Name them. ix) Write a note on Nucleus or Endoplasmic reticulum.	2016
ii) Write a note on Mitochondria or plastids	2015

Section C

5. Explain the chemical composition, structure and functions of plasma membrane.	2025
4. (OR) Explain the structure and composition of Cell Wall.	2024
7. Explain the chemical composition of Plasma membrane. Also describe its functions in regulating cell's interaction with environment.	2023
4. (b) Describe Fluid Mosaic Model of plasma membrane. (Diagram not require)	2022
5. With the help of Fluid Mosaic Model write the structure, properties and function of the plasma membrane.	2016

Chapter 4: BIOENERGETICS

Section B

x) Why ATP is known as common energy currency of living system?	2025
xi) What is the molecular formula of chlorophyll 'a' and 'b'?	
xiii) How C_4 Plants are different from C_3 plants?	
iii) Write a note on any one of the following: * CAM Plants * C_4 Plants	2024
iv) Why photorespiration considered as wasteful process?	
2. ix) Why ATP is called common energy currency of living system?	2023
2. (iv) What is ATP?	2022
2. (xi) Name first stable compound of Krebs cycle.	
3. (vi) Write a note on oxidative phosphorylation	
3. (viii) Describe role of water in light reaction.	
2. i) Why is ATP called Biological currency of energy?	2021
2. (ix) Why is ATP called the Currency of Energy?	2019
2. (x) Write the names of reactants and products of Photosynthesis.	
3. (viii) Draw a chart showing the efficiency of Food chain.	
3. (v) What is fermentation? Also name its types.	2018
2. (xv) Which compound receives CO_2 during dark reaction?	2017
2. (xvi) Name only Two types of fermentation.	
3. (iv) Why ATP called energy currency?	
xiv) Write a note on Oxidative phosphorylation.	2016
xv) Write the economic importance of fermentation.	
xvi) Write a note on Alternative mechanism of CO_2 fixation.	

iii) State the role of water OR chlorophyll in photosynthesis.	2015
x) State four important events take place during light dependent reaction of photosynthesis.	
Section C	
3. Explain light independent reaction of photosynthesis.	2025
3. —OR— Explain oxidation of Pyruvic acid and kreb's cycle.	
4. Describe Cyclic and Non-cyclic phosphorylation during light reaction.	2024
4. Describe oxidative degradation of Pyruvic Acid.	2023
4. (OR) Explain the light dependent reaction with reference to Electron transport chain and formation of ATP.	
5. (a) Describe the Glycolysis Pathway.	2022
6. Describe the light dependent reactions of photosynthesis.	2021
5. Define Cellular Respiration. Elucidate oxidative degradation of Pyruvic acid.	2019
6. Define Photosynthesis. Describe light reaction.	2018
7. With the help of a flow chart, describe the breakdown of glucose into pyruvate during glycolysis.	2017
5. OR Define a photosynthesis. Describe the light dependent reaction of photosynthesis	2016
5. Describe the process of Glycolysis. Also draw flowchart	2015

Chapter 5: ACCELLULAR LIFE

Section B

i) Write a note on Lytic cycle OR Lysogenic cycle.	2025
ix) Write a note on Viroids OR Prions.	
xiv) Write the stages of HIV life cycle.	
v) What are the symptoms of AIDS?	2024
2. viii) Classify Virus on the basis of Capsid.	2023
2. (v) Define Viroids.	2022
2. ii) Name any two diseases caused by Bacteria in human beings	2021
2. (ii) Name any two diseases caused by Virus.	2019
3. (iv) Differentiate between Lysogenic and Lytic cycle.	
3. (v) Write note on transmission of HIV virus.	
2. (xvi) What are Prions?	
2. (iv) What is Bacteriophage?	2018
3. (iv) Draw diagrammatic lytic cycle of bacteriophage	

2. (vi) Name any two diseases caused by virus.	2017
3. (iii) Draw a labeled diagram of Bacteriophage virus.	
vii) Write a note on transmission of HIV	2016

Chapter 6: PROKARYOTES

Section B

vi) Describe the classification of bacteria on the basis of shapes.	2025
vi) What are the various methods of control harmful bacteria?	2024
vii) Differentiate between Gram positive and Gram negative bacteria.	
2. (ii) Name types of bacteria on the basis of shape.	2022
3. (ii) Write salient features of Cyanobacteria.	
2. iv) Name three methods of genetic recombination found in Bacteria.	2021
3. iv) Write any four features of Cyanobacteria.	
3. vi) Classify Bacteria on the basis of flagella.	
2. (vii) Write only the name of the methods of Genetic recombination in Bacteria.	2019
2. (xiii) Draw a neat and labelled diagram of a Bacterial cell.	
3. (iii) Write the salient features of Cyanobacteria.	
2. (iv) Name the phases of Bacterial growth.	
3. (vii) Write four measures to control bacteria.	2018
2. (xv) What is Heterocyst ?	
2. (iii) How is bacterial cell wall different from plant cell wall?	2017
2. (vi) What is plasmid?	
3. (viii) How are bacteria classified on the basis of flagella?	
iii) Write the salient features of cyanobacteria.	2016
xiii) Methods of Genetic recombination in Bacteria.	
xiv) How bacteria is classified on the basis of Flagella.	2015

Section C

4. Describe the sources of nutrition in bacteria.	2025
6. What are Bacteria? Describe the structure of bacteria with the help of labelled diagram.	2024

Chapter 7: PROTOCTISTS AND FUNGI

Section B

viii) Differentiate between any One of the following: *Prokaryote and Eukaryote	2025
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viii) How prokaryotes are different from eukaryotes?	2024
ix) What features allow fungi to survive in all environments?	
2. iv) Name various divisions of fungi with one example each.	2023
2. (iii) Name various divisions of Fungi.	2022
2. (iii) Why is Deuteromycota called Fungi imperfect?	2019
3. (ix) Discuss the diversity of Kingdom Protoctista.	
2. (xiv) Name four groups of Fungi.	
2. (iii) Give two economic importance of Fungi.	2018
2. (x) What are lichens?	
2. (iv) Name various types of Ascocarp	2017
2. (vii) What are Mycorrhiza?	
3. (ii) Name only 4 groups of fungi with their reproductive structures.	
i) Name four groups of Fungi with their reproductive organs.	2016
xi) write note on lichen.	2015
Section C	
6. (b) Describe and draw the life cycle of Zygomycota fungi	2022
5. Draw and describe the life cycle of Zygomycota.	2021
5. Draw and describe the life cycle of Ulva or Fern.	2018
6. Draw and describe the life cycle of Zygomycota or Ascomycota fungi.	2017
3. OR Describe the structure and reproduction in Ulva with reference to Isomorphic alternation of generation.	2015

Chapter 8: DIVERSITY AMONG PLANTS

Section B

x) Give botanical names of any two of the following: * Tulsi * Amaltas * Mulethi * Mako	2024
xi) Write three steps in evolution of seed.	
xii) Why cold treatment is necessary for germination of seed in some plants?	
xiii) Differentiate between Monocotyledon and Dicotyledon.	
xiv) Define Cymose Inflorescence.	
2. i) According to Webbed Limby, how many-veined leaves evolved?	2023
2. (viii) Name any three subdivisions of Tracheophyta.	2022
2. (ix) Name various classes of Bryophyta.	
2. (vii) Write botanical names of any two of the following:	

* Wheat * Rice * Red Pepper * Rose 3. (iii) Describe any one of the following: * Evolution of Seed * Evolution of Leaf 3. (vii) Draw any one of the following: * Marchantia Thallus	
2. (iii) Define Heterospory. 2. (vi) Name only the sub-divisions of Tracheophyta. 2. (vii) Why is Anthocerotae considered an advance class of Bryophyta? 3. (i) Draw well labelled diagram of any one of the following: * T.S of Marchantia Thallus 3. (ii) Why are Bryophytes called the amphibious plants. 3. (iii) Discuss the steps involved in evolution of seed.	2021
2. (v) Write the botanical names of any two of the following: * Rice * Potato * Almond 2. (viii) What is Double fertilization? 2. (xii) Write the names of five groups of Vascular plants. 3. (vii) Write note on any one of the following: * Evolution of Leaf * Evolution of Seed	2019
2. (vii) Write the botanical names of any two of the following: (a) Tobacco (b) Edible pea (c) Rose (d) Rice 2. (xiv) Mention three steps of evolution of seed in plants. 3. (ii) Trace the evolution of leaf in plants. 3. (iii) Give the outline of the classification of Kingdom Plantae.	2018
2. (viii) Define Heterogamy. 2. (xii) Write the botanical names of any two of the following: * Wheat * Apple * Tomato * Amaltas 2. (xiii) Name different parts of a Carpel. 3. (i) Name only the sub-division of Tracheophyta. 3. (ix) Draw a labeled diagram of any one of the following: (ii) T/S of Marchantia thallus	2017
iv) Write the botanical names of any four of the following: * Brinjal * Tomato * Pear * Sweet pea * Rice vi) Name major groups of vascular plants with examples. vii) Write a note on the Evolution of Leaf or Seed.	2016
ix) Write down the botanical name of any four of the following. a) Rice b) Potato c) Tomato d) Wheat e) Tamarind xiii) Draw the labeled diagram of any one of the following: * L.S Male Cone of Pinus * L.S of Ovule of an anghiospermic plants xv) Draw an outline of classification of kingdom of plantae	2015
Section C	

3. Describe the life cycle of fern OR Flowering plants.	2024
3. Draw and describe the life cycle of Moss or Fern.	2023
4. (a) Draw and describe life cycle of Moss.	2022
4. Describe the life cycle of fern. (Diagram no required)	2021
6. [OR] With the help of diagram, describe the life cycle of Moss.	2019
4. Describe the life cycle of Fern	2016

Chapter 10: FORMS AND FUNCTIONS IN PLANT

Section B

ii) Define Halophytes. Write two examples of halophyte plants. iii) Describe the role of Sulphur OR phosphorus in Plants. vii) Why carnivorous plants use insects as food? xii) Define any One of the following: *Annual ring *Turgor pressure *Parenchyma	2025
2. ii) Write the role of any one macro nutrient in plant growth. 2. iii) How plants cope with High temperature? 2. v) Enlist any four adaptations in xerophytes to cope with water shortage. 2. vi) Write a note on different pathways for water to enter into Xylem. 2. vii) Name structural components of xylem and phloem. x) How annual rings are formed? 2. xi) Name various groups of plants on the basis of osmotic adjustment. 2. xii) Mention the role of Apical meristem.	2023
2. (xii) Define Ascent of Sap. 3. (v) Define Facilitated Diffusion 3. (iv) Mention the deficiency symptoms of Nitrogen in Plants.	2022
2. ix) Define Imbibition OR Osmosis. 3. vii) Mention any two factors which affect the opening and closing of stomata.	2021
2. (i) Name the pathways available for water to enter the Xylem. 3. (i) State Root Pressure Theory. 3. (vi) Define any two of the following: *Plasmolysis *Diffusion *Water potential	2019
2. (viii) Give two symptoms of nitrogen deficiency in plants. 3. (viii) Why do insectivorous plants trap insects while they also prepare carbohydrates by Photosynthesis ? 2. (ix) Define facilitated diffusion.	2018

2. (ix) Name three pathways of water absorption in plants.	2017
2. (xiv) Define Ascent of Sap	
viii) Describe the deficiency symptoms of Nitrogen or Potassium in Plants.	2016
x) What are the three different pathway available in root for water to enter Xylem?	
xi) Define any two of the following: Plasmaolysis b) Osmosis d) Active transport	
xii) Define Transpiration. Describe its various types in Plants.	
v) Describe the importance and deficiency symptoms of nitrogen in plants.	2015
xii) Explain source – Sink Movement.	
xvi) define any two of the following *Plasmolysis*Diffusion *Water potential*Imbibitions	
Section C	
6. Describe source-sink movement. How pressure flow hypothesis explain the mechanism of translocation of food?	2025
5. Define Ascent of Sap. Describe the cohesion and tension theory of water movement through xylem.	2024
5. What are growth regulators? Give their names and explain the role of Auxins?	2023
6. How does E. Munch hypothesis explain the translocation of food in plants?	
7. Discuss Transpiration pull theory.	2021
4. Define Transpiration. Describe the mechanism of Ascent of sap.	2018
5. OR Describe the mechanism of Ascent of Sap.	2015

MISCELLANEOUS QUESTIONS

QUESTIONS	CH	YEAR
3. v) Distinguish between any one of the following:		2021
* Angiosperm and Gymnosperm	8	
* Alcoholic Fermentation & lactic acid fermentation	5	
v) Distinguish between any one type:		2016
a) Angiosperm and Gymnosperm	8	
b) Photosynthesis and Respiration	13	
vi) Distinguish any one of the following:		2015
Aerobic Respiration and Anaerobic Respiration	13	
Angiosperm and gymnosperm	8	